

REPTILE SOFT TISSUE SURGERY

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1. INSTRUMENTATION

Microsurgical instruments

For medium size and small reptiles, microsurgical instruments are preferred. They should be a standard length (usually 6–7 inches long), have rounded handles and have small working tips. Many microsurgical instruments are also counter balanced and shaped to fit the notch between the base of the thumb and the index finger. When performing surgery, the hands rest firmly on the operating table. Both hands are positioned in the same manner as when we write, and the instruments are manipulated with the fingers only. The most important instruments in addition to the standard surgical instruments are the micro forceps, micro scissors, and micro needle holders. Other instruments can be added to the set, such as micro hemostats, vascular clamps, and others. DeBakey thumb forceps are atraumatic vascular forceps that can be used to handle delicate tissues.

Radiosurgery

Radiosurgery utilizes high frequency radiowaves to cut with little heat damage to the adjacent tissues. The Surgitron Dual Frequency 120 (Ellman International, Inc., Hewlett, NY, USA) employs 4.0-MHz offers monopolar/bipolar applications with foot-pedal/finger switch control for cut and coagulate. The ground is a plate with an antenna and does not even have to be in contact with the patient. It has both bipolar and monopolar settings. The monopolar cutting needles are mostly used for dissection, while the bipolar forceps are used for sealing blood vessels. The Harrison bipolar tip have one arm of the tip is bent at a 45° angle and are used for cutting avian skin. Radiosurgical units should not be used in the presence alcohol and carefully around paper, and plastic because are flammable material.

Magnification and illumination

Magnification is recommended to handle small and delicate tissues in small reptiles. The ideal surgical loupes system would be comfortable; provide magnification of 3–5X; have lenses with a large depth of field so more than one plane is in focus; have a focal distance appropriate to the surgeon's working distance, minimizing stresses on the neck and back (ergonomic); and allow the surgeon to look around the lenses to accomplish tasks that do not need magnification. Surgitel system (General Scientific Corporation, Inc. Ann Arbor, MI, USA) provides lenses that meet these criteria. Illumination with a focal light source is recommended. Lens-mounted lights direct the light where the surgeon needs keeping it in the field of view when the head is moved, but focal lights are also available separate from loupes.

Soft-tissue retractors

The soft-tissue retractors recommended are the smallest sized Balfour and Gelpi retractors for larger patients, and the Bennett avian retractor and the Doolen avian retractor (Sontec Instruments, Inc.) for smaller patients.¹ The Lone Star Retractor (Lone Star Medical Products, Inc., Stafford, TX, USA) is available in a variety of configurations but the rectangle, and it is likely the most versatile retractor used in avian surgery.

Vascular clips

Hemostatic clips, such as Hemoclips (Hemoclips and Samuels Hemoclip Applier, RICA Surgical Products, Inc.) are made of metal (steel, titanium, or tantalum) and serve the same purpose as ligatures.¹ They are available in various sizes. They are quick and easy to apply. It is recommended that the length of the clip be 2–3 times the diameter of the vessel. Right-angled appliers are also available and are especially useful for applications deep in the body cavity of small patients.

Sutures

Synthetic monofilament suture material that is rapidly absorbed (e.g. (Monocryl)) should be used in reptiles. Chromic gut, polyglyconate (Maxon), polyglactin 910 (Vicryl), and polyglecaprone (Monocryl) were evaluated histologically 7 days after laparoscopic sex determination in loggerhead sea turtles (*Caretta caretta*).² Results indicated that polyglecaprone and polyglyconate caused the least tissue reaction of the 4 suture types examined. These synthetic monofilament absorbable suture materials caused significantly less crust formation and panniculus inflammation than chromic gut and polyglactin 910. In a more recent study, Polydioxanone suture (PDS), PDS/Triclosan, polyglecaprone (Monocryl), Monocryl/Triclosan, polyglactin 910 (Vicryl 910), chromic gut, monofilament nylon, and surgical steel) and cyanoacrylate tissue adhesive in hatchling ball pythons (*Python regius*) over a period of 90 days.³ All suture materials caused chronic inflammatory responses that were significantly higher than the negative control 90 days after suture implantation. No sutures were completely absorbed by the end of the study period, suggesting that absorption times are prolonged compared with mammals. Despite previous reports that reptiles may be unable to break down chromic gut, there was histologic evidence of fragmentation of these sutures. Interestingly, there was also evidence that before complete absorption, prolonged absorption may lead to suture extrusion. The fewest and finest sutures possible should be used, and efforts should be made to minimize tissue trauma to decrease the risk of wound infection.

2. PATIENT PREPARATION

The skin is prepared with routine cleansings with chlorhexidine alternating with sterile saline. The use of excessive amounts of water, alcohol, and chlorhexidine may result in hypothermia. Chlorhexidine is more commonly used because it has residual activity, has a broad spectrum of activity, causes fewer skin reactions, and is not inactivated by organic material. Saline is recommended instead of alcohol because has proven to be as effective in preventing infection when alternated with the disinfectant solution, and because alcohol use will result in a greater heat loss.

Clear plastic drapes (Veterinary Transparent Surgical Drape, Veterinary Specialty Products, Inc., Shawnee, KS, USA), allow the patient to be monitored more closely, for example, for respiratory rate and character. These drapes are placed in addition of the quarter drapes around the patient and the biggest drape to cover the table.

3. SURGERY OF THE SKIN

Skin incision and closure

Skin incision is best performed between the scales whenever possible. Skin closure is best accomplished with an everting suture pattern such as horizontal or vertical mattress.⁴ Skin staples are also applicable to reptiles, since they are strong and cause eversion of the skin, which allows the incised edges to be maintained in apposition.

⁴The everting suture pattern leaves a raw edge exposed, and a light dressing or liquid bandage can be applied. Reptiles generally do not traumatize the suture lines. Suture removal is usually recommended 4-6 weeks after suture placement.⁴

Aural abscess removal

Most common in chelonians. The reptile is placed in sternal recumbency, with the head and neck extended. A ventral, semicircular incisional flap, or a circular incision in extreme cases, is made over the distended tympanum.⁵ The skin is gently retracted to allow the insertion of a curette around the edge of the caseous material. The caseous material abscess is removed carefully with a small curette. The cavity is flushed copiously with sterile saline. The site is left open and flushed daily.⁵

4. COELIOTOMY

Snakes

The precise surgical site is determined using anatomy references,⁶ palpation and diagnostic imaging. The snake is placed in lateral recumbency. A permanent marker is used to mark the surgical site. In general, the incision is made between the first and second rows of lateral scales. This ensures that the incision is positioned laterally and off the floor when the recovered snake is moving. Dissection is continued through the muscle layer, entering into the coelom between the ribs and ventral musculature. The coelom is accessed after transecting several soft tissue layer, and extensive fat bodies can difficult visualization. A two-layer closure is routine, with the muscle layer closed using absorbable suture in a simple interrupted or continuous pattern. Skin closure is routine

Lizards

Most lizards are placed in dorsal recumbency, but some laterally compressed species (e.g. chameleons) may be better placed in left or right lateral position.⁵ The head and body are supported, and the legs are secured with tape.

For the paramedian approach, the skin is incised parallel to, but lateral from, the ventral midline. The abdominal musculature is incised. The coelomic membrane is very thin and is easily breached using cotton-tipped applicators. Care is required to identify a distended bladder and avoid accidental rupture upon entry into the coelom. In small lizards, there is no need to suture the coelomic membrane or thin abdominal musculature, only the skin need be closed. Coeliotomy closure in larger lizards should be in two layers, with the muscle closed using a simple interrupted or continuous pattern with absorbable suture. Skin closure is routine. The advantage of this technique is clear avoidance of the large midline abdominal vein, which is a confluence of the bilateral pelvic veins.⁵ Between the cranial margin of the pubis extending approximately one fourth of the distance toward the umbilicus and cranial to the xiphoid, no midline abdominal vein is present.⁴ The disadvantage, the incision through abdominal musculature may produce more postoperative discomfort.⁵

For the midline approach requires an incision started at the umbilical scar and extended craniad.⁵ Careful incision through the skin and underlying linea alba reveals the abdominal vein that should be preserved and gently retracted laterally. If the ventral abdominal vein is cut, and ligated, the blood is redirected back through the pelvic veins into the bilateral renal veins and directly into the vena cava. A simple interrupted or continuous pattern, using absorbable suture material is used to close the linea alba. Skin closure is routine. The advantages of this technique are equal accessibility to both sides of the coelom and less muscle trauma; however, greater surgical skill is required to preserve the abdominal vein.⁵ The urinary bladder (if present), colon or fat bodies may obstruct visualization of other visceral structures, and need gentle manipulation and retraction using cotton tipped applicators.⁵

For the lateral approach, in laterally compressed species (e.g. chameleons), a skin incision is made parallel to the ribs. The intercostal muscles are incised, and if needed a rib or two might be transected. Closure is routine.

Chelonians

Depending on the species and the organ of interest, the two major approaches are transplastron and prefemoral. Transplastron coeliotomy: provides full access to the coelom (e.g. gastrointestinal tract, reproductive tract, bladder). The chelonian is placed in dorsal recumbency. The four incision (two lateral, one cranial and one caudal) over the plastron are planned considering the degree of access required, position of the heart, position of two parallel abdominal veins and the presence of plastron hinges. The saw is angled in for each cut. After osteotomy, the section of bone is elevated from the underlying abdominal musculature with the back of the scalpel handle or, more appropriately, a periosteal elevator. Once the bone is reflected and the abdominal musculature visualized, two large veins are identified, one on either side of the midline. One should avoid damaging these sinuses; however, if necessary, they may be ligated.⁴ A ventral midline incision should be performed through the easily distensible coelomic membrane to allow access to the coelomic cavity. The ventral midline should be closed with a simple continuous pattern with an absorbable material. The bone flap is carefully cleaned and placed back into the osteotomy site. The flap is sealed in place with a waterproof rapid-setting epoxy resin. The placement of bone wax in the incision site to prevent epoxy from dripping into the coelom.⁴

Prefemoral coeliotomy: provides limited access to the coelom (e.g. reproductive tract, bladder). The chelonian is placed in lateral or laterodorsal recumbency, the pelvic limb is secured caudad to expose the prefemoral fossa. A craniocaudal incision is made in the middle of the fossa. Blunt dissection cranial to the sartorius and ventral to the iliacus muscles reveals the coelomic aponeurosis of the transverse and oblique abdominal muscles.⁵ This coelomic aponeurosis is fibrous and some force may be required to incise the membrane. Closure is routine.

5. SURGERY OF THE GASTROINTESTINAL SYSTEM

Esophagostomy tube placement

Most common in chelonians but also in lizards. The esophagostomy tube should be premeasured before placement. The tube length should be measured and marked from the lateral surface of the neck to the midbody of the chelonian.⁷ To avoid accidental incision of an artery or vein, a hemostat can be used as a guard. The hemostat should

be inserted through the oral cavity and into the esophagus and displaced laterally. A small incision should be made through the skin and esophagus with a #15 blade scalpel. The hemostats should be manipulated through the incision, so that they can be used to grasp the feeding tube. The hemostats and the feeding tube should be withdrawn through the oral cavity. The tube should only be withdrawn to the point of the landmark. Once withdrawn, the feeding tube should be redirected down the esophagus into the stomach.⁷ The tube should be sutured to the skin at the level of the incision with nonabsorbable sutures in a chinese finger trap, and the extended length of the tube secured to the carapace with tape or cyanoacrylic glue.

Gastrotomy/ Enterotomy

Removal of foreign body, trauma and neoplasia may require gastrotomy or enterotomy. The same principles for gastrointestinal surgery as in small animal apply in reptiles. If intestinal resection and anastomosis is performed, using a 6-0 to 10-0 absorbable monofilament suture on a one-fourth-circle atraumatic needle is recommended. Typically six to eight simple interrupted sutures are necessary for end-to-end anastomosis.

Cloacal prolapse repair

Minor cloacal prolapse are reduced and held in place with simple interrupted or mattress sutures on both sides of the vent while the underlying cause is treated. Major or recurrent cloacal prolapse may require cloacopexy. A ventral midline coeliotomy is performed and the cloaca identified. The cloaca is sutured full thickness or partial thickness with incisions created in the serosa into the ventral midline body wall. For the first method, the suture passes through one side of the body wall, through the full thickness of the cloaca and through the other side of the body wall in a simple interrupted pattern. For the second, incisions are created in the serosal surface of both the body wall and the cloaca, paramedian approximately 5-10 mm from the midline. Three or four sutures are placed between each side of the 2 incisions such that the serosal surfaces are sutured and the subserosal surfaces are opposed. In cases of severe cloacal and colonic necrosis, resection and end-to-end anastomosis are essential.⁵

6. SURGERY OF THE RESPIRATORY SYSTEM

Tracheal resection and anastomosis

Neoplasia (chondromas) may result in tracheal stenosis in ball pythons.^{8,9} The affected segment of the trachea may be resected. The trachea is approached as described for celiotomy in snakes on the basis of the radiographic location of the lesion. Snakes do not have a recurrent laryngeal nerve as is seen in mammals.⁴ The trachea is transected circumferentially a tracheal ring cranial and caudal to the affected segment with a No. 11 scalpel blade. The tracheal ends are approximated by preplacing simple interrupted sutures of 4-0 to 6-0 PDS on a tapered needle. The sutures are tightened individually to appose the tracheal ends. Tension-relieving sutures are placed if needed on the ventral and lateral aspects of the trachea, each encircling a tracheal cartilage proximal and distal to the anastomosis site. All suture knots are extraluminal. Snakes have C-shaped tracheal rings and care should be exercised not to damage the dorsal membrane when manipulating the structure.⁴ Closure of the surgical site is routine.

7. SURGERY OF THE REPRODUCTIVE SYSTEM

Ovariectomy

In lizards that are not reproductively active, the ovaries are generally small. They are located adjacent to the renal veins and adrenal glands in the mid to caudal coelom. Ovary removal can be safely accomplished by first applying vascular clips along the mesovarium to ligate the small vessels before dissecting the gonad free. It is not necessary to remove the oviducts if they are normal.⁵ In cases of preovulatory ova stasis (POOS), the active ovaries are considerably enlarged and immediately obvious upon entering the lizard coelom as they resemble lusters of yellow–orange grapes. Each ovary is elevated to expose the suspensory ligament containing 3–8 large vessels that branch off the aorta and renal veins. Hemoclips or suture material are used to ligate these vessels, before the ovary is dissected free. The oviducts are usually small and involuted and do not need to be removed unless diseased.⁵

In chelonians is performed as part of dystocia surgery to prevent future breeding; a transplastron or bilateral prefemoral coeliotomy is required. The chelonian ovaries originate close to the ventrolateral aspect of the kidneys, but in mature females they are large and extend into the central coelom. The mesovarium is extensive and great care is required to ensure that the entire ovary has been isolated, before vascular clips are applied and the ovary removed.⁵

In snakes an incision extending for 20–25% of the snout-to-cloaca length would be required to expose both ovaries.⁵

Salpingotomy/salpingectomy

In cases of postovulatory egg stasis (POES) or dystocia, it is the thin oviducts full of eggs or fetuses that are

immediately obvious upon entry into the coelom of lizards and snakes. Multiple salpingotomy incisions can be made to remove the eggs or fetuses in an effort to maintain future breeding capacity.⁵

In most lizards, bilateral ovariosalpingectomy is recommended. The large, numerous blood vessels that supply each oviduct must be ligated with suture or vascular clips. Vascular clips and lasers greatly reduce surgery time and several vessels can often be clamped with a single clip. The oviducts are ligated close to their insertion with the cloaca using a circumferential transfixing ligature, and removed.

In snakes, the length of the oviducts tends to argue against salpingectomy. Instead, single or multiple coeliotomy and salpingotomy incisions are preferred, with efforts made to manipulate more cranial and caudal eggs out of the same incision.⁵

Chelonian oviducts are thicker than those of lizards and snakes. The surgeon often has the viable option of performing multiple salpingotomy incisions and preserving the reproductive status of the animal, or a unilateral or bilateral ovariosalpingectomy.⁵

Orchidectomy

Most common in lizards, rare in snakes and very difficult in chelonians. In lizards, the testes are generally located adjacent to the dorsal renal veins and adrenal glands, in the mid to caudal coelom. The testes are suspended by a short mesorchium from the dorsal body wall, within which the testicular artery and veins are located. The left adrenal gland is positioned between the left testis and the renal vein, and care must be exercised to identify and preserve the gland during orchidectomy. The right adrenal is below the

renal vein and is therefore less likely to be damaged.⁴ The testicle is retracted with the aid of a stay suture through the capsule. The testicle is gently retracted ventrally and a vascular hemostat or hemostatic clip is applied to the base of one testicle. The hemostat or clip must be applied parallel to the spine to avoid entrapping the renal vein. If possible, a second clip is applied just ventral to the first. One clip is applied in a cranio-caudal direction and the second clip in a caudo-cranial direction to incorporate the entire vascular supply. The base of the testicle is incised between the hemostat or clip and the ventrally applied clips with a scalpel blade. If there is remaining testicular tissue that protrudes through the hemostatic clip, should be removed or ablated with electrocautery or a laser. Residual testicular tissue may result in tissue hyperplasia and produce reproductive hormones. Before the coelomic cavity is closed, the vascular pedicles should be checked for residual hemorrhage.

Phallectomy

Chronic prolapse of the phallus (chelonians, crocodiles) or hemipenes (snakes, lizards) may require amputation. The reptilian copulatory organ does not contain an urethra; therefore circumferential or transfixion ligatures can be placed. In small reptiles, the phallus or hemipene are penetrated with absorbable 3-0 to 5-0 suture proximal to the affected tissue, and sutured with a transfixion suture pattern. The phallus or hemipene is cut distal to the ligature, and the stump is reintroduced in the cloaca or hemipene sac. In larger reptiles, after transection of the soft tissues, a continuous suture pattern can be placed to approach the mucosa.

8. SURGERY OF THE MUSKULOSKELETAL SYSTEM

Tail amputation

In lizards that perform autotomy, tail amputation can be most easily accomplished by simply snapping and twisting the tail through a fracture plane of a coccygeal vertebra.⁵ The muscle strands are trimmed and local antiseptic can be applied to the open wound. Even though autotomy is a natural phenomenon, anesthesia and aseptic preparation are mandatory.⁵ This author believes that surgical autotomy is the best method of amputation and required in tails that do not lend themselves to autotomy (e.g. proximal amputation of large mature lizards). The tail is amputated in a wedge fashion, with the flaps of the wedge on each side of the coccygeal vertebrae closing in a laterally compressed fashion (a near-normal anatomic apposition).⁴ The amputations may regrow some of the tail.⁴

Transmitter placement

The snakes are measured from the tip of the nose to the vent. The surgical site is approximately three-fourths of this distance from the nose. This places the transmitter just anterior to the gonads but caudal to the air sacs. A 4 to 5 cm skin incision is made between the second and third row of scales dorsal to the ventral scutes. The muscle layer is incised caudal to a rib or a hemostat is used to punch into the body cavity between two ribs. The rib is grasped at the cranial edge of the incision. The transmitter is pushed into the incision, below the rib and into the coelom with the antenna protruding from the incision and running cranially. A 24cm pipette tube is inserted under the skin from the incision and passed anteriorly. The antenna is threaded through this pipette. A small incision over the cranial tip of the pipette creates just large enough of a

hole from which to remove the pipette. The pipette is removed through the small incision and the antenna should remain in the subcutaneous space. The small incision is closed with a drop of surgical glue. The coelomic incision is closed in two stages. The muscle layer is closed using a continuous pattern and 3-0 or 4-0 absorbable monofilament suture and incorporating ribs to strengthen the closure. The skin is closed using an everting mattress pattern and the same 3-0 or 4-0 suture used for the muscle layer.

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